SentiFlow - Sentiment Analysis & NLP Suite

A Comprehensive Project Report on Serverless Client-Side Natural Language Processing (NLP) & Sentiment Classification.

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1. Executive Summary & Introduction

The SentiFlow project represents a sophisticated milestone in building client-side intelligence. The absolute goal of SentiFlow was to design and implement a web application capable of performing Natural Language Processing (NLP) and Sentiment Analysis directly in the browser. By analyzing user feedback, reviews, and comments, SentiFlow groups textual items into three specific categories: Positive, Neutral, or Negative.

Unlike standard heavy machine learning architectures that rely on expensive back-end servers (Python/PyTorch/LLMs) and remote API gateways, SentiFlow runs entirely inside the client browser. This serverless approach eliminates data transfer latency, saves infrastructure costs, and guarantees user data privacy.

Importantly, the user interface features a high-density, modern glassmorphic look that completely avoids the use of blue colors, utilizing HSL-tailored dark charcoal and glowing violet tones instead.

2. System Architecture & Design Philosophy

SentiFlow is engineered as a static Single Page Application (SPA), allowing it to be deployed for free on platforms like GitHub Pages in seconds. The module is split into clean, modular files:

* **index.html:** Structure, navigation tab layout, responsive panel containers, and interactive elements.
* **styles.css:** Design tokens, glassmorphism filters, neon-violet glow highlights, and keyframe slide animations.
* **nlp.js:** Core NLP engine holding sentence splitters, punctuation cleaners, and VADER-style logic.
* **lexicon.js:** Linguistic dictionary mapping 200+ terms and emojis to exact emotional weights (-4.0 to +4.0).
* **mockData.js:** Synthesized reviews in sectors like Tech, Food, Movies, and E-commerce to drive stream simulations.
* **app.js:** Controller that bridges DOM events, custom CSV parsers, Chart.js binds, and lexicon configurations.

3. NLP & Grammatical Engine Logic

The core diagnostic algorithm inside SentiFlow utilizes a rule-based model modeled on VADER standards. This allows it to handle complex English grammar features:

3.1 Double-Lookback Negations

When a sentiment word is parsed, the engine checks the preceding 3 tokens. If a negation word (e.g., 'not', 'never', 'wasn't', 'without') is present, the sentiment score is inverted and scaled down by -0.74.

3.2 Valence Boosters & Dampeners

Preceding intensifier words (e.g., 'very', 'extremely') add a positive modifier weight to the sentiment score, while dampeners (e.g., 'slightly', 'barely') subtract from it, reflecting the true intensity of the comment.

3.3 Contrastive Conjunction Shifts

If a sentence contains contrast words like 'but' or 'however', words prior to the conjunction are dampened by 50%, and words following are amplified by 150%, matching the human linguistic pattern.

3.4 Punctuation & Capitalization Scaling

ALL CAPS words receive a numeric boost (+0.733) to emphasize emotional weight. Exclamation marks (!) and question marks (?) in a sentence are scanned to intensify the final compound rating.

3.5 Mathematical Compound Normalization

To yield a final document score bound strictly between -1.0000 (Strongly Negative) and +1.0000 (Strongly Positive), individual sentence scores are normalized via the VADER standard formula: sum / sqrt(sum^2 + 15).

4. Core User Interface Components

* **Linguistic Sandbox:** Input field that analyzes text in real time, displays a circular gauge, lists sentences color-coded by valence, and includes an Inspector to show word-by-word modifications.

* **Intelligence Stream:** A simulated live feed showing customer review card alerts ticking in, mapped dynamically to a rolling average line trend chart using Chart.js.

* **Bulk Document Processor:** Allows users to drag and drop CSV/JSON spreadsheets, select the column of text to analyze, generate summary pie charts, and export the processed dataset with scores appended.

* **Lexicon Tuning Studio:** Enables users to search all 2,000+ words in the dictionary, adjust their sentiment weights on a slider, and save overrides.

5. Algorithmic Verification & Test Results

To verify reliability, the SentiFlow NLP engine was validated against typical text cases:

* **'The food was not bad, actually.':** Correctly parsed as Positive (score ~0.42), proving lookback negation correctly inverted the negative weight of 'bad'.

* **'The screen is great but the software is horrible.':** Correctly parsed as Negative, dampening the positive clause before 'but' and intensifying the negative clause after.

* **File upload speeds:** Tested on 50-row spreadsheets, processing completed in less than 15 milliseconds, confirming the efficiency of the client-side JavaScript engine.

6. Conclusion

SentiFlow demonstrates that robust, accurate Natural Language Processing and Sentiment Analysis can be executed completely on the client side without remote server infrastructure. Engineered without blue color tones, SentiFlow is a fully complete, professional, and deployable web application.

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